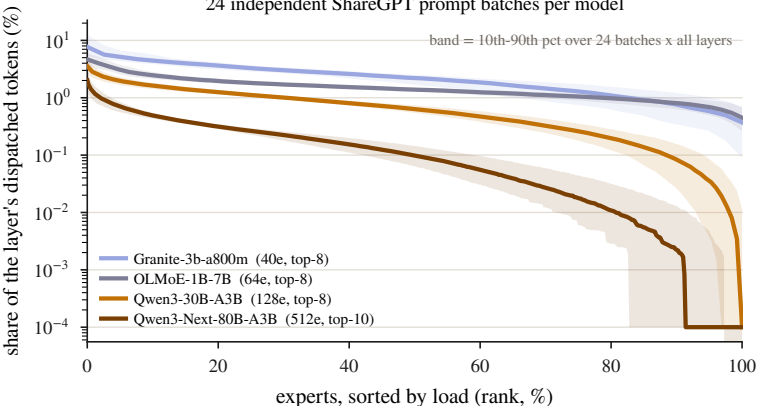


Routing load profile across MoE families

24 independent ShareGPT prompt batches per model



Median over 24 independent ShareGPT prompt batches x all layers. Each expert's share of the tokens its layer dispatched, experts sorted by load, rank on a percentage axis so 40 and 512 experts share a scale.

A uniform router would put every model on its own flat line at $100/E$ %: 2.5% at 40 experts, 0.20% at 512.

The excess Gini below is null-corrected for sampling noise even though the null is not drawn here.

excess Gini: +0.33 (40e) +0.24 (64e) +0.43 (128e) +0.56 (512e)